



Possible explanations for the upward trend in mental distress among adolescents in Norway from 2011 to 2024

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ABSTRACT

Background: An upward trend in self-reported mental distress among adolescents has been documented in Norway and several other countries, yet the causes remain unclear. This study aims to identify potential explanations for this trend by testing hypothesized factors using repeated cross-sectional data.

Methods: We analyzed responses from 979,043 Norwegian adolescents, collected across 1417 municipality level surveys between 2011 and 2024. Eight explanatory factors with prior evidence of association with mental distress were examined: family financial difficulty, evenings spent at home, physical inactivity, bullying victimization, dissatisfaction with parents, social media use, cannabis use, and school dissatisfaction. Following a published epidemiological framework, we assessed changes in prevalence, virulence (strength of association), and the degree to which each factor statistically explained the trend in mental distress.

Results: Mental distress increased significantly for both sexes over time. Increased school dissatisfaction accounted for a large share of the observed trend. Time spent on social media, also explained a part of the trend. Spending more evenings at home and cannabis use contributed modestly. Other factors, including financial difficulty, bullying victimization, physical inactivity, and dissatisfaction with parents, showed limited explanatory value.

Conclusions: The rising trend in adolescent mental distress in Norway appears most plausibly linked to growing school dissatisfaction, with a possible contributory role of social media use. These findings underscore the need for further research into school-related factors and adolescent digital experiences, as well as preventive measures.

Mental distress refers to symptoms of common mental disorders, typically including depressed mood and worry, and is widely assessed with self-report questionnaires in adolescent population based surveys (Olsen et al., 2006). Evidence suggests an upward trend in mental distress among young people in many countries, including Norway, over the past decades (Boer et al., 2023; Keyes and Platt, 2024; Krokstad et al., 2024; McGorry et al., 2024; Nilsen et al., 2024; Potrebny et al., 2024). Concurrently, there has been an increase in mental healthcare utilization and treatment (McGorry et al., 2024). The causes of these trends remain poorly understood, highlighting the need for more research to inform policy and to plan interventions.

A wide range of explanations have been proposed for the upward trend in mental distress, including global developments such as climate change, the spread of neoliberalism, growing intergenerational economic inequality, and diminished future prospects (McGorry et al., 2024). Negative developments related to school have also been proposed, including increased bullying (Keyes and Platt, 2024), school disengagement, and school stress (Armitage et al., 2024; Boer et al., 2023; Cosma et al., 2020). Changes in adolescents' leisure time activities have also been suggested as explanations, such as reduced free play (Haidt, 2024), and increased use of social media (Keyes and Platt, 2024; Twenge, 2017). It has also been suggested that recent adolescent cohorts

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have been shielded from discomfort, which has led to less resilience (Haidt and Lukianoff, 2018). Furthermore, according to ‘the prevalence inflation hypothesis’, increased awareness about mental health may have led to a rise in self-reported symptoms because of improved recognition of mental health issues, but also by overinterpretation of common psychological experiences (Foulkes and Andrews, 2023).

Despite widespread attention in the media, surprisingly few empirical studies have explicitly attempted to explain upward trends in mental distress (Armitage et al., 2024). Determining the causes of trends is challenging. Experimental designs are typically infeasible for practical and ethical reasons, and while notable quasi-experimental studies exist (e.g., Braghieri et al., 2022), most studies that attempt to explain trends rely on analysis of existing repeated cross-sectional data. Keyes and Platt (2024) recently proposed an epidemiological framework with a useful set of criteria to identify plausible explanations for the upward trend in mental distress. Their first criterion is an individual level association, ideally a causal relation, between the explanatory factor and mental distress. Secondly, the prevalence of the explanatory factor should increase over time, or, thirdly, the strength of its association with mental distress should increase over time. The latter is referred to as increased virulence, implying that the same ‘dose’ of the explanatory factor becomes more harmful over time. Due to stronger increase in mental distress among girls than boys in the USA, Keyes and Platt also added that the increase in virulence or prevalence of potential explanatory factors should be more pronounced among girls.

The aim of the current study is to examine if the upward trend in mental distress among Norwegian adolescents can be attributed to specific hypothesized explanations, and thereby to narrow the range of plausible explanations. We analyzed data from a large collection of youth surveys conducted in Norwegian municipalities since 2010. Because the most plausible explanations are individual-level predictors of mental distress, (Rutter, 1995), we began by identifying candidate factors where there is evidence of prospective individual level associations. The identification of factors was based on their relevance to adolescent mental health and their availability across data sets. The selection was informed by the authors’ knowledge of the field, ad-hoc literature review, but not by observed trends in the data. The identified factors were (with studies providing evidence of prospective individual level associations in brackets): subjective financial hardship (Frankham et al., 2020; Kiely et al., 2015), spending leisure time at home rather than with friends (Hunter et al., 2024; Hutten et al., 2021; Liu et al., 2020), lack of physical activity (Duncan et al., 2022; Morgan et al., 2024; Wu et al., 2018), bullying victimization (Smokowski et al., 2014; Stapinski et al., 2014; Wei et al., 2023), dissatisfaction with parents (Chen and Harris, 2019; Dhondt et al., 2019), time spent on social media (Colasante et al., 2024; Fitzpatrick et al., 2023; Lee et al., 2022; Riehm et al., 2019; Viner et al., 2019), cannabis use (Gripe et al., 2024; Patton et al., 2002; Sawyer et al., 2024), and school dissatisfaction (Lester et al., 2013; Shochet and Smith, 2014; Zhang et al., 2023).

Some of these factors have also been associated with shifts in adolescent mental distress over time. Increased parent-child conflict and increased school disengagement may partly have explained the increase in mental distress among 15-year olds in Scotland between 1987 and 2006 (Sweeting et al., 2010). Increased cannabis use may have contributed to an increase in mental distress among Norwegian 16-17 year-olds between 1992 and 2002 (von Soest and Wichstrøm, 2014). A decrease in bullying victimization may have prevented an increase in mental distress in the Netherlands between 2005 and 2017 (De Looze et al., 2020). Higher mental distress among British 16/17-year olds born in 2001 compared to 1958 can partly be explained by the association between indicators of low socioeconomic status and mental distress becoming stronger over time (McElroy et al., 2023). However, others have argued that there is only limited empirical evidence that trends in adolescent mental distress can be explained by changes in leisure time socialization with friends, lack of physical activity, and time spent on social media (Armitage et al., 2024).

Following the criteria suggested by Keyes and Platt (2024), we examined changes in mental distress from 2011 to 2024. Next, we assessed trends in eight factors with prior evidence of prospective associations with mental distress: family financial difficulty, the number of whole evenings spent at home per week, (lack of) physical activity, bullying victimization, dissatisfaction with parents, time spent on social media, cannabis use, and school dissatisfaction. We then examined change in virulence for all eight factors over the study period. Finally, we tested if holding constant changes in the prevalence of the explanatory factors would alter the slope of the upward trends in mental distress.

1. Method

1.1. Data

We analyzed data from Ungdata, which is a collection of municipal repeated cross-sectional surveys in Norway that have used identical procedures and core questionnaires yearly since 2010 (see <https://www.ungdata.no/English/>). The goal is that each of Norway’s municipalities (357 as of January 1, 2024) conduct a survey every three years. The survey data is used to monitor living conditions for adolescents in lower secondary school (grades 8 to 10; ages 12–16) and upper secondary school (grades 11 to 13; ages 16 to 19) in each participating municipality. The collection of municipal surveys is also used as a national monitoring system with annual assessments. The surveys cover a broad set of themes including school, family and living conditions, mental and physical health, and leisure time activities. In all survey years, data were collected in early spring using e-questionnaires administered in classrooms. The average response rate for the municipal surveys was 83.7 % (SD = 8.7). In this study, we used data from all Ungdata surveys completed between 2011 and 2024, but 2020 was excluded because of disruption by the COVID-19 pandemic. The surveys conducted in 2010 were not included because of reverse ordering of the mental distress response options. The analytical dataset consisted of data from 1417 local surveys. The number of respondents varied between survey years; the smallest was 11,971 respondents in 2011, the largest was 136,540 in 2021. The total number of respondents in the analytic sample was 979,043 (489,702 boys and 489,341 girls).

1.2. Measures

1.2.1. Outcome variable: Mental distress

Mental distress was measured with six questionnaire items from Kandel and Davies’ Depressive Mood Inventory (Kandel and Davies, 1982). Using a four point scale ranging from “Not been affected at all” to “Been affected a great deal”, respondents indicated their agreement with the following assertions: 1) “Felt that everything is a struggle”, 2) “Had sleep problems”, 3) “Felt unhappy, sad or depressed”, 4) “Felt hopelessness about the future”, 5) “Felt stiff or tense”, and 6) “Worried too much about things”. A detailed assessment of the Norwegian version of this measure concluded that the instrument was measurement invariant over time (Nilsen et al., 2024). Composite scores were computed by calculating the mean of these six items (Cronbach’s alpha was 0.89).

1.2.2. Explanatory factors

The explanatory variables were measured with questionnaire items that were developed for Ungdata or for a series of preceding youth surveys called Young in Norway (von Soest and Wichstrøm, 2014).

Financial difficulty was measured with the item ‘Financially, has your family been well off, or badly off, over the past two years?’ The response options were ‘We have been well-off the whole time’; ‘We have mostly been well-off’; ‘We have neither been well-off nor in financial difficulty’; ‘We have mostly been in financial difficulty’; and ‘We have been in financial difficulty the whole time’. These response options were coded 1 to 5.

The second item was ‘Think back over the last week ... How many times have you spent the whole evening at home?’ The response options were ‘None’ (coded 1); ‘1 time’ (coded 2); ‘2–5 times’ (coded 3); ‘6 times or more’ (coded 4). These response options were coded 1 to 4; higher scores indicate a greater number of whole evenings spent at home.

The questionnaire included three items about physical activity, introduced by ‘How often do you participate in the following activities’: ‘Train or compete with a sports club’, ‘Go to the gym’, and ‘Other kinds of organized physical activity (dance, martial arts, etc.)’. There were six response options: ‘Never’, ‘Rarely’, ‘1–2 times a month’, ‘1–2 times a week’, ‘3–4 times a week’, and ‘At et 5 times a week’. These response options were coded 1 to 6.

Bullying victimization was measured with the item ‘Are you yourself subjected to bullying, threats, or exclusion by other young people at school or during your free time?’. The response options were ‘Yes, several times a week’ (coded 6), ‘Yes, around once a week’ (coded 5), ‘Yes, around once a fortnight’ (coded 4), ‘Yes, around once a month’ (coded 3), ‘Almost never’ (coded 2), and ‘Never’ (coded 1). Higher scores indicate greater bullying victimization.

Satisfaction with parents was measured with an item taken from a list of questions that were introduced by ‘How happy or unhappy are you with various aspects of your life?’, and included the item ‘Your parents’. The response options were ‘Very unhappy’, ‘Slightly unhappy’, ‘Neither happy nor unhappy’, ‘Quite happy’, and ‘Very happy’. These response options were coded 1 to 5.

A questionnaire item about social media read ‘Think about what you do on a normal day. How much time do you spend on the following things: Social media (Facebook, Instagram, etc.)’. The response options were ‘No time’, ‘Less than 30 min’, ‘30 min - 1 h’, ‘1–2 h’, ‘2–3 h’, and ‘More than 3 h’, and were coded 1 to 6.

Cannabis use was measured with an item that read ‘How many times have you done any of the following things over the past year: Used hash or marijuana’. The response options were ‘Never’, ‘Once’, ‘2–5 times’, ‘6–10 times’, and ‘11 times or more’, and were coded 1 to 5.

School dissatisfaction was measured with one questionnaire item from a list introduced by ‘Do you agree or disagree with the following statements about your situation at school’. The item read ‘I enjoy school’. The response options were ‘Totally agree’, ‘Somewhat agree’, ‘Somewhat disagree’, and ‘Totally disagree’. The response options were coded 1 to 4, and higher scores indicated greater school dissatisfaction.

All but two of the variables were measured from 2011 to 2024. Social media use and school dissatisfaction were not measured until a questionnaire revision in 2014. Descriptive statistics for the study variables are presented in Table 1.

Each municipality’s degree of geographic centrality was included in the dataset, based on Statistics Norway’s centrality index. The centrality index ranges from 1 (high) to 6 (low). Municipalities with high centrality have higher density of goods and service providers, more variation in work opportunities, and higher rates of tertiary education (Høydahl, 2017). Along with school grade (8th = age 12/13 ... 13th = age 18/19), it was used as a covariate in the analyses.

1.3. Analysis

The analyses were performed in Stata version 18 (StataCorp LLC, 2023). We examined change in mean scores over time in mental distress, individual mental distress items, and potential explanatory factors for mental distress from 2011 to 2024 with linear regression models. In these models, time (year) was entered as a continuous variable. For illustrative purposes, we also estimated a linear regression model where year was coded as dummy variables.

Changes over time in the association between mental distress and each explanatory factor (i.e., changes in virulence) were estimated by regressing mental distress on the explanatory factor, time (year), and the time (year) by explanatory factor interaction term.

Finally, the explanatory factors were examined one-by-one in separate regression models where mental distress was the outcome variable, and time (year) was a continuous predictor. We examined if the estimated trend in mental distress would change when each of the explanatory factors were held constant in separate regression models.

Differences between the estimates were tested with Wald-tests based on the ‘seemingly unrelated estimation’ postestimation command (‘suest’) in Stata.

All models were estimated separately for boys and girls. We used standard errors clustered at years and municipalities to account for any dependency between adolescents who attended school in the same municipality in the same year. Also, all models included dummy-variables for school grade with 8th grade as the reference category, and dummy-variables for geographic centrality with most central as the references category, due to potential age and centrality differences in mental distress (Abebe et al., 2016; Brunborg et al., 2024), and potential differences in the explanatory factors. The alpha level for statistical significance was set at $p < 0.001$.

2. Results

2.1. The upward trend in mental distress from 2011 to 2024

Mean mental distress estimates for Norwegian adolescents for each year from 2011 to 2024 are shown in Fig. 1. Scores were higher for girls than for boys at all time points. There were fluctuations, but the mean scores increased over time for both sexes. The estimated annual linear increase in mental distress was $b = 0.014$ (95 % CI: 0.012, 0.015; $p < 0.001$) for boys, and $b = 0.016$ (95 % CI: 0.014, 0.018; $p < 0.001$) for girls. A combined model for boys and girls with a time by sex interaction term indicated a slightly stronger increase for girls ($b = 0.003$ (95 % CI: 0.001, 0.004); $p < 0.001$). The estimated standardized effect size (Cohen’s d) comparing the start and the end of the study period was $d = 0.25$ for boys and $d = 0.26$ for girls.

Linear changes in all individual mental distress items were also calculated. As shown in Table 2, there were linear increases in all six types of distress for girls, and all items except the ‘felt that everything is a struggle’ item for boys. This item also showed the weakest increase for girls. The strongest increase, for both boys and girls, was for the item

Table 1
Descriptive statistics.

Variable	Range	Boys	Girls	Data collection years
		M (SD) N	M (SD) N	
Mental distress	1–4	1.81 (0.71) 443,476	2.31 (0.80) 461,247	2011–2024 excl. 2020
Financial difficulty	1–5	1.79 (0.90) 470,455	1.91 (0.93) 474,700	2011–2024 excl. 2020
At home all evening	1–4	2.91 (0.88) 452,228	2.95 (0.81) 468,624	2011–2024 excl. 2020
Physical activity	1–6	4.30 (1.55) 450,221	4.11 (1.43) 467,078	2011–2024 excl. 2020
Bullying victimization	1–6	1.61 (1.12) 468,857	1.63 (1.12) 477,272	2011–2024 excl. 2020
Satisfaction with parents	1–5	4.46 (1.07) 439,646	4.37 (1.12) 463,801	2011–2024 excl. 2020
Time on social media	1–6	3.83 (1.52) 391,945	4.71 (1.34) 410,122	2014–2024 excl. 2020
Cannabis use	1–5	1.21 (0.76) 461,343	1.12 (0.53) 474,291	2011–2024 excl. 2020
School dissatisfaction	1–4	1.55 (0.75) 410,658	1.60 (0.76) 412,989	2014–2024 excl. 2020

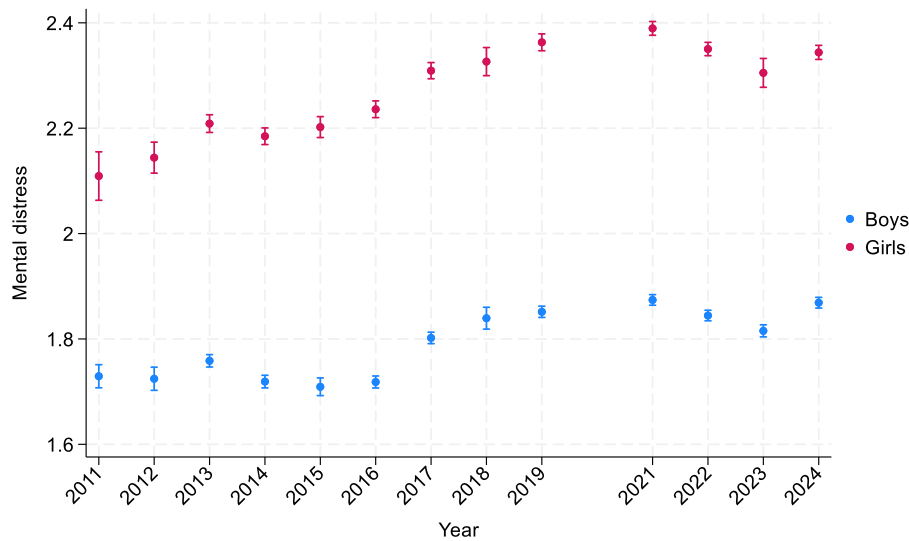


Fig. 1. Mean mental distress scores from 2011 to 2024 for Norwegian adolescents. Estimates are adjusted for differences in school grade and geographic centrality. Confidence intervals are based on standard errors adjusted for municipality year clusters.

Table 2

Average annual change for individual mental distress items, from 2011 to 2024 among Norwegian adolescents.

	Boys			Girls		
	<i>b</i>	SE <i>b</i>	β	<i>b</i>	SE <i>b</i>	β
'Felt that everything is a struggle'	0.001	0.001	0.001	0.007	0.001	0.007
'Had sleep problems'	0.017	0.001	0.018	0.018	0.001	0.018
'Felt unhappy, sad or depressed'	0.017	0.001	0.017	0.011	0.001	0.011
'Felt hopelessness about the future'	0.015	0.001	0.015	0.016	0.001	0.016
'Felt stiff or tense'	0.011	0.001	0.011	0.020	0.001	0.021
'Worried too much about things'	0.021	0.001	0.020	0.026	0.001	0.024

Note: Estimates adjusted for school grade and geographical centrality. *b* = unstandardized regression coefficient; β = y-standardized regression coefficients. Bold denotes that *b* is statistically significantly different from zero at the $p < 0.001$ level.

'worried too much about things'.

2.2. Trends in potential explanatory factors

Next, we examined if there were upward trends in the potential explanatory factors over the studied time-period. Means of standardized scores (z-scores) for each year are shown in Fig. 2, and the estimates for linear annual changes using both original and standardized mental distress variables are presented in Table 3.

For both sexes, there were linear increases in spending evenings at home, spending time on social media, cannabis use, and school dissatisfaction. The increase was greater for time on social media and school dissatisfaction than for spending evenings at home, and the increase in cannabis use was rather small. Bullying victimization remained stable over time, while financial difficulty decreased over time and physical activity increased. For boys only, satisfaction with parents also increased over time, but this change was small.

2.3. Change in the virulence of the potential explanatory factors over the studied time-period

Next, we examined if the associations between mental distress and each of the explanatory variables changed over time, that is, we

examined change in virulence. Table 4 shows the main effect for each explanatory variable and each of the interactions with time, i.e., the annual change in virulence. For both sexes, all the potential explanatory factors were cross-sectionally associated with mental distress in the expected direction. As indicated by the standardized estimates, the strength of associations varied from physical activity (weakest) to school dissatisfaction (strongest).

For boys, the virulence increased for financial difficulty, spending whole evenings at home, and school dissatisfaction. For girls the virulence increased for financial difficulty, and for spending whole evenings at home, however, decreased virulence was observed for bullying victimization, cannabis use, time on social media, and school dissatisfaction.

2.4. Change in time trend for mental distress over the studied time-period when holding constant the effects of potential explanatory factors

A series of linear regression models were run to examine if holding constant the effect of each of the explanatory variables would reduce the estimated upward trend in mental distress. The results are presented in Table 5.

For boys, holding constant any change over time in spending all evening at home resulted in statistically significant change in the annual increase estimate ($\chi^2(1) = 258.58, p < 0.001$), but the estimate only changed marginally (from $b = 0.013$ to $b = 0.012$). Holding constant time on social media also resulted in statistically significant change in the annual increase in mental distress estimate ($\chi^2(1) = 911.50, p < 0.001$): The estimate for annual increase in average mental distress from 2014 to 2024 dropped from $b = 0.016$ to $b = 0.011$. Holding school dissatisfaction constant also resulted in a significant change in the annual increase estimate ($\chi^2(1) = 786.20, p < 0.001$). The estimate for annual increase in average mental distress dropped from $b = 0.016$ to $b = 0.001$, a level that was not significantly different from zero.

For boys, holding constant any change over time in any of the other potential explanatory factors (financial difficulty, physical activity, bullying victimization, dissatisfaction with parents and cannabis use) did not reduce the estimated upward trend in adolescent mental distress.

Similar results were observed for girls. Holding constant the increase in spending whole evenings at home resulted in a small but significant change in the estimated annual increase in mental distress ($\chi^2(1) = 210.01, p < 0.001$): from $b = 0.016$ to $b = 0.014$. A small reduction in the annual increase in mental distress estimate was also observed after holding constant cannabis use ($\chi^2(1) = 116.47, p < 0.001$). The estimate

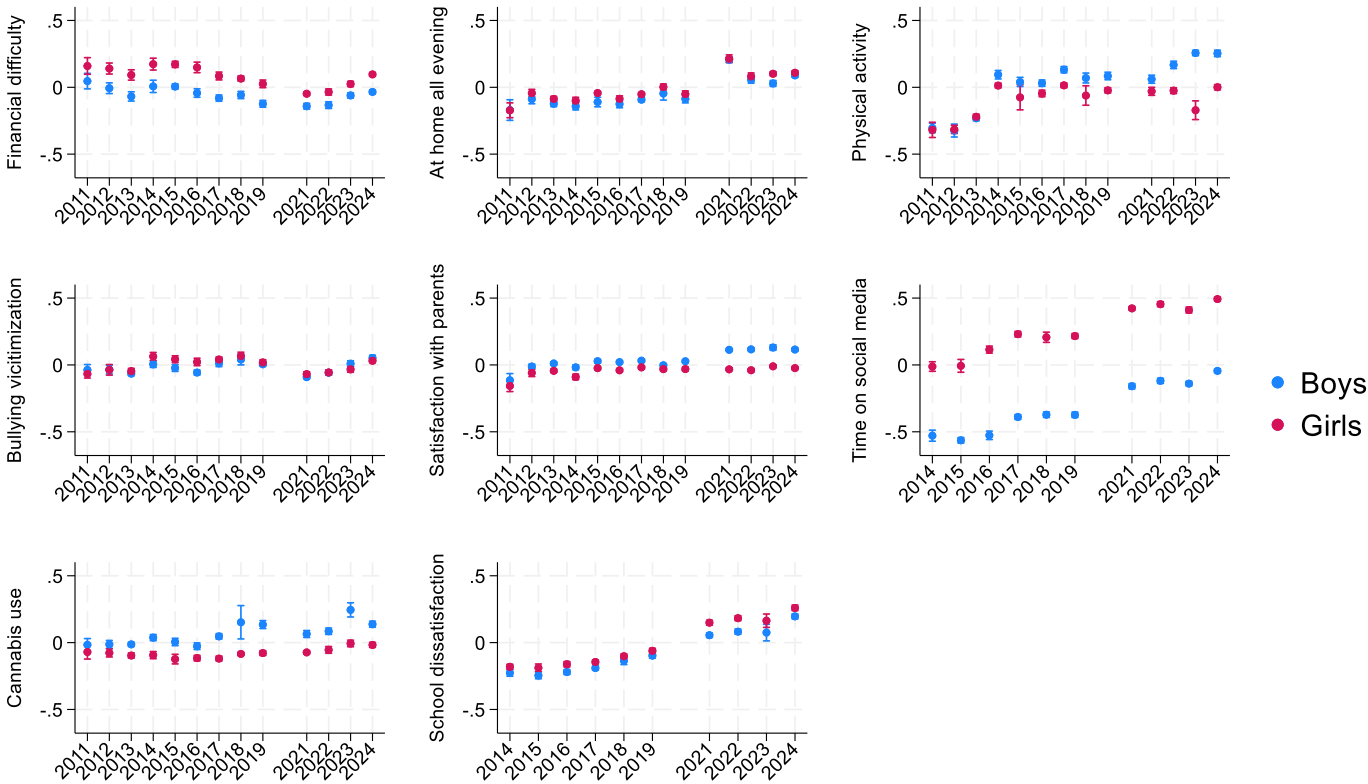


Fig. 2. Estimated mean z-scores for potential explanatory factors for mental distress across the study period. Estimates are adjusted for differences in school grade and geographic centrality. Confidence intervals are based on standard errors adjusted for municipality year clusters.

Table 3
Average linear change per year for potential explanatory factors for increased risk of mental distress.

Time period		Boys			Girls		
		<i>b</i>	SE <i>b</i>	β	<i>b</i>	SE <i>b</i>	β
Financial difficulty	2011–2024	–0.006	0.001	–0.006	–0.011	0.001	–0.012
At home all evening	2011–2024	0.022	0.001	0.026	0.020	0.001	0.023
Physical activity	2011–2024	0.045	0.003	0.030	0.016	0.003	0.011
Bullying victimization	2011–2024	0.003	0.001	0.002	–0.003	0.001	–0.003
Satisfaction with parents	2011–2024	0.015	0.001	0.013	0.002	0.001	0.002
Time on social media	2014–2024	0.084	0.002	0.056	0.079	0.003	0.053
Cannabis use	2011–2024	0.008	0.001	0.012	0.006	0.001	0.009
School dissatisfaction	2014–2024	0.037	0.001	0.048	0.041	0.001	0.053

Note: Estimates adjusted for school grade and geographical centrality. Bold denotes that *b* is statistically significantly different from zero at the $p < 0.001$ level. β denotes estimates using y-standardization.

Table 4
Change in virulence over time for each of the potential explanatory factors.

Time period		Boys				Girls			
		Main effect			Annual change in virulence	Main effect			Annual change in virulence
		<i>b</i>	SE <i>b</i>	β		<i>b</i>	SE <i>b</i>	β	
Financial difficulty	2011–2024	0.179	0.001	0.23	0.006 <0.001	0.208	0.002	0.24	0.003 <0.001
At home all evening	2011–2024	0.091	0.002	0.11	0.004 <0.001	0.116	0.002	0.11	0.004 0.001
Physical activity	2011–2024	–0.029	0.001	–0.06	–0.000 <0.001	–0.046	0.002	–0.08	–0.001 <0.001
Bullying victimization	2011–2024	0.192	0.001	0.30	–0.001 <0.001	0.226	0.002	0.31	–0.003 <0.001
Satisfaction with parents	2011–2024	–0.130	0.001	–0.20	–0.001 <0.001	–0.167	0.002	–0.23	–0.001 <0.001
Time on social media	2014–2024	0.059	0.001	0.13	–0.001 <0.001	0.106	0.001	0.18	–0.002 <0.001
Cannabis use	2011–2024	0.151	0.004	0.16	0.002 0.001	0.185	0.005	0.12	–0.005 0.001
School dissatisfaction	2014–2024	0.382	0.002	0.40	0.004 0.001	0.458	0.003	0.44	–0.007 0.001

Note: Estimates adjusted for school grade and geographical centrality. *b* = unstandardized regression coefficient. Standard errors are adjusted for municipality clusters. β = y-standardized regression coefficient. Bold denotes statistical significance at the $p < 0.001$ level.

Table 5

Mean change per year in composite mental distress scores for Norwegian adolescent boys and girls estimated from linear regression models with different sets of covariates.

Model #	Covariates	Time-period	Boys		Girls	
			<i>b</i>	SE <i>b</i>	<i>b</i>	SE <i>b</i>
1	Age, Centrality	2011–2024	0.013	0.001	0.016	0.001
2	Age, Centrality, Financial difficulty	2011–2024	0.015	0.001	0.018	0.001
3	Age, Centrality, At home all evening	2011–2024	0.012	0.001	0.014	0.001
4	Age, Centrality, Physical activity	2011–2024	0.015	0.001	0.017	0.001
5	Age, Centrality, Bullying victimization	2011–2024	0.013	0.001	0.017	0.001
6	Age, Centrality, Satisfaction with parents	2011–2024	0.016	0.001	0.017	0.001
7	Age, Centrality, Cannabis use	2011–2024	0.013	0.001	0.015	0.001
8	Age, Centrality	2014–2024	0.016	0.001	0.015	0.001
9	Age, Centrality, Time on social media	2014–2024	0.011	0.001	0.006	0.001
10	Age, Centrality, School dissatisfaction	2014–2024	0.001	0.001	–0.004	0.001

Note: *b* = unstandardized regression coefficient. Standard errors are adjusted for municipality clusters. Estimates in bold are statistically significantly different from zero at the $p < 0.001$ level. Models 8–10 cover a shorter time period because Time on social media and School dissatisfaction were not measured between 2011 and 2013.

changed from $b = 0.016$ to $b = 0.015$. A larger reduction in the annual increase in mental distress was observed after holding time on social media constant: From $b = 0.015$ to $b = 0.006$ ($\chi^2(1) = 764.04$, $p < 0.001$). Holding constant change over time in school dissatisfaction resulted in an even larger reduction in the annual increase estimate, from $b = 0.015$ to $b = -0.004$ ($\chi^2(1) = 843.17$, $p < 0.001$) a level that was not significantly different from zero.

For girls, holding constant any change over time in any of the other potential explanatory factors (financial difficulty, physical activity, bullying victimization, and dissatisfaction with parents) did not reduce the estimated upward trend in adolescent mental distress.

3. Discussion

This study provides insight into potential explanations for the upward trend in adolescent mental distress observed in Norway between 2011 and 2024. Using Keyes and Platt's (2024) epidemiological framework, we show that three out of eight considered potential explanations emerge as plausible contributors to the upward trend for both girls and boys. These are increases in school dissatisfaction, time on social media, and spending whole evenings at home. Increased cannabis use also emerged as a plausible but minor contributor for girls only. The remaining four factors emerged as unlikely contributors for both sexes: increases in financial difficulty, physical inactivity, bullying victimization and dissatisfaction with parents.

We now discuss our findings in relation to the three criteria in this framework. First, all the potential explanatory factors considered in our study have previously been shown to be longitudinally associated with mental distress. This was further corroborated in our study by significant cross-sectional associations between these factors and mental distress. This fulfills the first of Keyes & Platt's criteria.

Secondly, four of the factors exhibited increases over time, fulfilling Keyes and Platt's second criterion. These included spending whole

evenings at home, time on social media, cannabis use, and school dissatisfaction. Of these, spending whole evenings at home (for both sexes) and school dissatisfaction (for boys only) showed increased virulence over time. These results support these factors as plausible contributors to the upward trend in adolescent mental distress. In contrast, perceived financial difficulty, physical exercise, dissatisfaction with parents, and bullying victimization showed no increases over time. With the exception of perceived financial difficulty, none of these factors showed increased virulence over time either.

We were able to add a criterion, namely that a factor is more likely to be explanatory if holding it constant over time reduces or attenuates the estimated upward trend in mental distress. For both sexes, holding constant time on social media and school dissatisfaction significantly attenuated the estimated upward trend. Holding constant change over time in evenings spent at home also resulted in small changes in the estimate for both sexes. For girls, a small reduction also occurred when holding constant cannabis use.

A summary of the study results is presented in Table 6. Based on our analysis, four of the eight potential explanatory factors seem unlikely: Changes in perceived financial difficulty, physical inactivity, bullying victimization and dissatisfaction with parents. By the same token, possible explanations for the upward trend in mental distress include increased school dissatisfaction and increased time spent on social media, and to a lesser extent increased number of evenings spent at home and increased cannabis use.

Increased school dissatisfaction emerged as the most prominent potential explanation for the upward trend in mental distress. This is in line with previous studies suggesting associations at the individual level (Lester et al., 2013; Shochet and Smith, 2014; Zhang et al., 2023), and studies that have found associations between increases in mental distress over time and increases in school disengagement (Sweeting et al., 2010) and schoolwork pressure (Cosma et al., 2020). Increasing school dissatisfaction over time may stem from multiple sources, including heightened academic and social pressure, peer-related challenges, and misalignment between adolescent expectations and educational experiences. It should be noted that the causal direction between school dissatisfaction and mental distress is not obvious (Lester et al.,

Table 6

Summary of study findings.

Potential explanatory factors	Associated with mental distress?	Increased over the same time period? ^a	Increase in virulence?	Holding constant reduced estimated increase in mental distress?
Perceived financial difficulty	Yes	No	Yes	No
At home all evening	Yes	Yes	Yes	Yes, but small effect
Physical inactivity	Yes, but small effect	No	No	No
Bullying victimization	Yes	No	No	No
Dissatisfaction with parents	Yes	No	No	No
Cannabis use	Yes	Yes	No	Yes, but small effect for girls only
Time on social media	Yes	Yes	No	Yes
School dissatisfaction	Yes	Yes	Yes, but for boys only	Yes

Note.

^a From 2011 to 2024 for all potential explanatory factors except Time on social media and School dissatisfaction, where increases were assessed from 2014 to 2024.

2013; Shochet and Smith, 2014; Zhang et al., 2023), and some evidence suggest a bidirectional relation (Leurent et al., 2021). However, a review of studies that have examined the relation between factors closely related to school dissatisfaction (e.g., school connectedness, school engagement, school belonging, school climate) in longitudinal studies concluded that higher school connectedness can predict lower mental distress among adolescents (Raniti et al., 2022).

Increased time spent on social media also emerged as a possible explanation for the upward trend in mental distress. To our knowledge, no previous studies have examined directly if time spent on social media can impact trends in adolescent mental distress (Armitage et al., 2024), however, a review that combined evidence from multiple sources concluded that social media is a plausible explanation for the increase in mental distress (Keyes and Platt, 2024). Our findings align with previous cross-sectional and longitudinal individual level studies suggesting that high levels of social media engagement are associated with increased mental distress (Appel et al., 2020; Colasante et al., 2024; Fitzpatrick et al., 2023; Lee et al., 2022; Riehm et al., 2019; Viner et al., 2019). Our finding is also in line with evidence that increased national level general internet use was associated with increased national-level psychological complaints in the Health Behavior in School-aged Children study between 2002 and 2008 (Boer et al., 2023). There is also evidence that restricting social media use can improve mental health in college students and adult samples, though effect sizes are typically small (Burnell et al., 2025; Thimm-Kaiser and Keyes, 2025; Thrul et al., 2025). Proposed mechanisms include increased social comparison, online bullying and reduced face-to-face social interactions. The findings from restriction-based studies are, however contested, and other studies do not necessarily find the expected positive effects of restricting access to smartphones or social media (Ferguson, 2025; Goodyear, Randhawa, et al., 2025). It should also be noted that a number of studies have concluded that there is no prospective relation between social media use and mental distress (Beeres et al., 2021; Boer et al., 2021; Coyne et al., 2020; Ferguson et al., 2025; Plackett et al., 2023; Steinsbekk et al., 2023). There is also evidence of a bi-directional relationship (Frison and Eggermont, 2017; Vidal et al., 2020). Our finding that time spent on social media can partially explain the increase in mental distress rests on the assumption that the association is causal at the individual level. Further investigation is required to establish if this is indeed the case. For instance, it is possible that adolescents who experience mental distress may increase their social media use to distract themselves from negative emotions.

There is currently debate in many countries about restricting the use of social media among adolescents, such as banning mobile phones from classrooms and schools (Selwyn and Aagaard, 2021). Given the limited understanding of how social media use and its restriction can affect mental distress (Burnell et al., 2025; Thimm-Kaiser and Keyes, 2025), the potential benefits of such interventions remain uncertain. It has also been argued that blanket bans on smartphones or social media are likely to be ineffective and potentially harmful, and that the focus should instead be on a multifaceted strategy involving digital literacy, safe design, and technology-free spaces (Goodyear, James, et al., 2025).

We could not rule out that an increase in spending evenings at home has played a role in the increase in mental distress over time. To our knowledge, this has not been studied before, but our finding is in line with studies showing that loneliness and social isolation can be detrimental to mental health (Hunter et al., 2024; Hutten et al., 2021; Liu et al., 2020). We also found that the virulence increased for spending evenings at home, which is in line with the finding that loneliness became more strongly associated with mental distress over time in a Norwegian study (Krokstad et al., 2024). Adolescents who experience mental distress may have a tendency to avoid socialization with peers, hence the causal direction is not obvious in this case either.

Our results indicated that increased cannabis use may have played a small role in increased mental distress, for girls only. This highlights the potential role of substance use in the context of adolescent mental

health, and the finding echoes results from a previous Norwegian study where increased cannabis use may have contributed to a previous increase in mental distress (von Soest and Wichstrøm, 2014). Both studies assume that cannabis use causes mental distress, and as there is also evidence of cannabis use to cope with mental health (Brodbeck et al., 2007), this may be a strong assumption. As for spending more time at home, we can more safely conclude that increased cannabis use, at best, can only explain a small portion of the upward trend in mental distress.

Interestingly, our results suggest that changes in perceived family financial situation, physical activity, bullying victimization, and satisfaction with parents cannot explain the upward trend in mental distress. The overall family financial situation improved slightly, while the virulence appeared to increase, however holding constant financial difficulty over time did not affect the estimated increase in mental distress over time.

Physical activity increased rather than decreased alongside the upward trend in mental distress, and the virulence did not change over time, making it an unlikely explanatory factor. This is in contrast to a recent Norwegian study (Krokstad et al., 2024) that did report small decreases in physical activity during a partially overlapping time period, and increased differences in mental distress according to physical activity for girls (i.e., increased virulence). However, the magnitude of these findings suggests that changes in physical activity may only have played a small role in the increased mental distress over time.

Bullying victimization largely remained stable, and it became less virulent for girls, which also makes this an unlikely explanatory factor for the increase in mental distress. A study from Wales also found that an increase in emotional problems among adolescents could not be explained by bullying, despite increases in both face-to-face and cyberbullying during the same time period (Anthony et al., 2024). Krokstad et al. (2024) reported slight increases in bullying from 2006 to 2019 alongside slight decreases in virulence, which suggests that increased bullying has not played a major role in the increased trends in mental distress.

Satisfaction with parents also did not decrease over time (it increased slightly for boys) and holding it constant did not affect the estimated increase in mental distress. A previous study found that increased parent-child conflict may partly have explained the increase in mental distress among Scottish adolescents between 1987 and 2006 (Sweeting et al., 2010), which does not align with our findings. The studies are not directly comparable, however, due to country differences, different time periods, and the fact that we measured satisfaction with parents and not child-parent conflict.

While our analyses examined the contribution of the explanatory factors independently, we acknowledge that some factors may be connected through underlying patterns of association or dependence. For example, increased social media use may be linked to reduced physical activity, spending time at home, or fights with parents. Alternatively, time spent at home can lead to increased social media use, social isolation, and reduced physical activity. Also, there might be an overlap between school dissatisfaction and social media use. Online conflicts can spill over into the school context, potentially increasing bullying or contributing to disengagement. Also, adolescents who are dissatisfied with school may turn to increased online activity to cope or escape. Future research should examine such interconnections more closely. Exploring interactions and potential mediation pathways is an important avenue for future research.

3.1. Strengths and limitations

A major strength of our study is the inclusion of large annual samples of adolescents, all with access to the same set of variables. Additionally, the individual surveys included in our study demonstrate strong response rates. Furthermore, we employed diverse analytical approaches to explore the feasibility of various factors in explaining the increases in mental distress over time.

Our study also has limitations. First, our analysis rests on several assumptions. As discussed, our analyses assume that the causal direction uniformly goes from each explanatory factor to mental distress. This strong assumption is the study's greatest limitation, and this has to be kept in mind when interpreting the results. The aim of the study was not to estimate causal effects, but rather to narrow the field of plausible explanations for the upward trend in mental distress.

Although we accounted for the main dependence structure of the data by clustering at the municipality and year of data collection, we cannot exclude the possibility that a small subset of individual respondents participated more than once, which could slightly bias the estimated standard errors and test statistics. However, given the very large sample size, we do not believe this has any substantive bearing on the main results and conclusions drawn.

All analyses were conducted with Norwegian data. Therefore, analysis using data from other countries is needed to assess the generalizability of our findings to other countries. All variables were measured using self-report with simple questionnaire items that give imperfect measurement, which can introduce bias and measurement error. This may particularly be important for the social media variable, where the top category was 3 or more hours per day. We estimated linear models to assess upward trends in both mental distress and the potential explanatory factors to aid interpretability, but we acknowledge that more flexible modelling of changes over time might have improved model fits. Finally, we only examined the potential explanations that were available in the dataset, which provides a limited array of explanatory variables. Other factors might explain the upward trend in mental distress.

4. Conclusion

This study provides insight into the factors that may have contributed to the upward trend in adolescent mental distress in Norway. Increased school dissatisfaction, increased social media usage, more evenings spent at home, and cannabis use were identified as potential explanatory factors, though their estimated effects differed. These results underscore the complexity of mental health trends and highlight the importance of considering multiple potential explanations. While the causal direction of these associations remains uncertain, the study raises questions for future research, particularly regarding the interplay between social media engagement and adolescent well-being. We are better suited to conclude that changes in perceived family financial situation, physical activity, bullying victimization, and satisfaction with parents are unlikely explanations of the upward trend in mental distress. Our findings can serve as a foundation for future studies aiming to provide stronger evidence for these factors as explanations for the rising rates of mental distress. Addressing these factors through interventions and further longitudinal studies is essential to mitigate the high levels of mental distress among adolescents.

CRedit authorship contribution statement

Geir Scott Brunborg: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sondre Aasen Nilssen:** Writing – review & editing, Writing – original draft, Validation, Investigation, Conceptualization. **Jens Christoffer Skogen:** Writing – review & editing, Writing – original draft, Validation, Investigation, Conceptualization. **Lasse Bang:** Writing – review & editing, Writing – original draft, Validation, Supervision, Investigation, Conceptualization.

Statement of ethics approval not needed

This study analyzed existing secondary data that had been previously collected and anonymized for research purposes. No new data were collected from human participants, and no identifiable personal information was accessed or used during the analysis. Therefore, ethical

approval for this study was not required.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT 4o in order to summarize text for the abstract and article highlights. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of interest statement

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Data availability

The authors do not have permission to share data.

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